Mechanism-Aware Prediction of Action Potential Initiation in Myelinated Axons under Extracellular Stimulation

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***Abstract*—Predicting the initiation site of action potentials (APs) in myelinated axons under extracellular stimulation is critical for understanding neural excitation and optimizing neuromodulation. Biophysically detailed models such as the McIntyre–Richardson–Grill (MRG) model provide accurate predictions but are computationally expensive. Here, we propose a mechanism-aware convolutional neural network (CNN) framework to efficiently predict AP initiation in myelinated axons. Based on MRG simulations of 400,000 stimulation scenarios, AP initiation was categorized into spatial gradient–driven and terminal depolarization–driven mechanisms. The framework first classifies the activation mechanism and then predicts the AP initiation site using dedicated CNN models. The CNN achieved 98.46% accuracy in mechanism classification and 99.59% accuracy in identifying initiation terminals. For spatial gradient–driven activation, the initiation site was predicted with a test-set mean absolute error of 1.21×10^{-2} and R^2 of 0.976. Compared with activating function and driving force methods, the proposed approach demonstrated substantially improved accuracy and robustness, providing a fast and mechanism-aware surrogate for axonal excitation prediction.**

***Index Terms*—extracellular electrical stimulation, action potential, myelinated axon model, neural network**

I. Introduction

Extracellular electrical stimulation is widely used in neuromodulation therapies such as spinal cord stimulation [1]–[3], deep brain stimulation [4], and peripheral nerve stimulation [5]. The neural response to an applied electric field is fundamentally determined by where and how action potentials (APs) are initiated along axons. Accurate identification of AP initiation sites is therefore essential for understanding neural recruitment mechanisms, optimizing stimulation strategies, and improving the efficacy and selectivity of neuromodulation.

In computational axon models, including the McIntyre–Richardson–Grill (MRG) model [6], [7], AP initiation under extracellular stimulation can generally be categorized into two distinct mechanisms [8] (Fig. 1). In the first mechanism, APs are initiated at internal nodes of the axon due to local variations in the spatial profile of the extracellular potential. This mechanism is commonly associated with the spatial gradient of the applied field along the axon and is often analyzed using the activating function (AF) [9], [10], defined as the second spatial derivative of the extracellular potential. In the second mechanism, APs are initiated at one of the axonal terminals as a result of direct depolarization at the boundary, even when the spatial second derivative of the extracellular potential along the axon is small or even negative. This terminal depolarization–driven activation is frequently observed in finite-length axons but cannot be explained by AF-based theories [8].

Biophysically detailed models such as the MRG axon model provide a rigorous framework for capturing both activation mechanisms and their resulting membrane dynamics [7]. However, simulating AP initiation using such models is computationally expensive, making them impractical for large-scale parameter exploration or real-time applications [11]. Consequently, simplified analytical approaches have been proposed to infer AP initiation sites directly from extracellular potential distributions. The AF method identifies candidate initiation nodes by locating the maximum of the activating function [12], [13], while extensions such as the driving force (DF) method incorporate weighted contributions from neighboring nodes to partially account for spatial interactions [14]–[16].

Despite their efficiency, AF- and DF-based methods suffer from fundamental limitations. They rely on predefined analytical assumptions, are sensitive to discretization and parameter selection [15], and are only applicable to spatial gradient–driven activation. In particular, these methods fail for potential distributions with constant second derivatives and cannot handle terminal depolarization–driven AP initiation.

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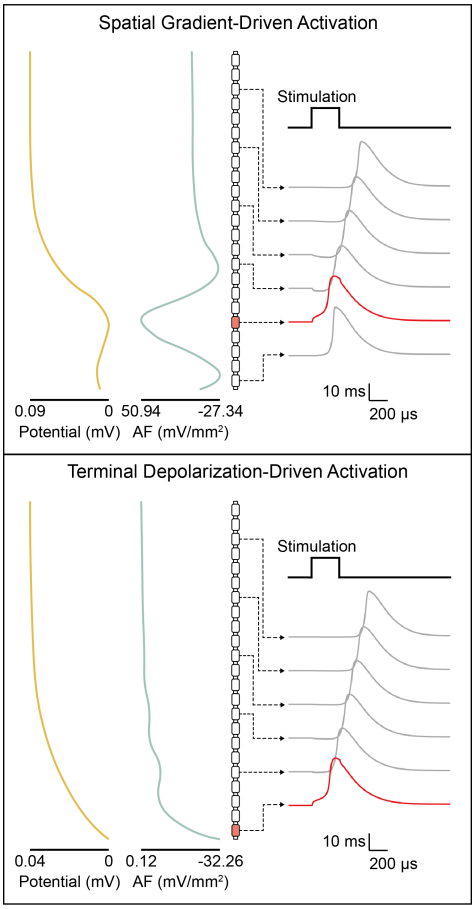


Fig. 1. Two mechanisms of action potential initiation.

Moreover, neither approach provides a unified framework capable of identifying the activation mechanism and predicting the initiation site simultaneously.

In this study, we propose a data-driven framework based on convolutional neural networks (CNNs) to predict AP initiation in the MRG axon model under extracellular stimulation. Leveraging a large-scale dataset generated from NEURON [11] simulations, the proposed CNN directly maps the extracellular potential distribution and axonal parameters to the AP activation mechanism and initiation site. To address the intrinsic heterogeneity arising from the two distinct activation mechanisms, we adopt a hierarchical prediction strategy: the network first classifies whether AP initiation is driven by spatial gradients or terminal depolarization, and then predicts the corresponding initiation location using task-specific subnetworks.

The proposed approach achieves accurate and robust prediction across a wide range of stimulation conditions and significantly outperforms traditional AF- and DF-based methods. By providing fast inference without explicit biophysical assumptions, this framework offers a practical surrogate for detailed axon simulations and enables efficient modeling and optimization of extracellular neuromodulation strategies.

II. Materials and Methods

A. Combined Datasets for Machine Learning

To enable data-driven prediction of action potential (AP) initiation sites, we employed a dataset consisting of 400,000 samples, which was originally constructed in our previous work [17]. Each sample includes the extracellular potential distribution along the axon, the axon diameter (5.7/7.3/8.7/10.0/11.5/12.8/14.0/15.0/16.0 μm), the axon length (ranging from 2 mm to 10 mm), and the stimulus pulse width (ranging from 100 μs to 1000 μs). Among these samples, 200,000 extracellular potential distributions were randomly generated, while the remaining 200,000 were obtained from finite element simulations using a personalized epidural electrical stimulation model [17]–[19]. For each sample, the axon diameter, axon length, and pulse width were independently and uniformly sampled within their respective ranges. The extracellular potential distribution was represented by a 1×100 vector constructed using interpolation, which was concatenated with three normalized scalar parameters (axon diameter, axon length, and pulse width) to form a 1×103 input vector for the neural network.

For all 400,000 samples, membrane potential responses of the MRG axon model under single-pulse extracellular stimulation were simulated using NEURON [11]. Model files of MRG axons were obtained from ModelDB with accession number 3810 [20]. The stimulus amplitude was set to 1.1 times the activation threshold to ensure reliable AP generation. The extracellular potential distribution was applied to all axonal nodes using the *play* method. Simulations were performed with a time step of 5 μs, and membrane potentials of all nodes were recorded over an 8-ms simulation window. The node at which the membrane potential first reached its maximum value was defined as the AP initiation node (Fig. 1). In cases where multiple nodes reached the maximum membrane potential simultaneously, the median index of these nodes was selected as the initiation node. The AP initiation site was then normalized by the total axon length and expressed as a relative position along the axon. If the AP was initiated at either the first or the last node, the activation was classified as *terminal depolarization–driven*. Otherwise, it was classified as *spatial gradient–driven*. Within the dataset, 43.1% of samples exhibited spatial gradient–driven activation, while the remaining 56.9% corresponded to terminal depolarization–driven activation.

The experimental procedures described in this paper purely involve computational modeling and biophysical simulations. No human subjects or animal models were involved in this study, and thus no institutional ethics committee approval was required.

B. Neural Network Construction and Training

The proposed architecture utilizes cascaded convolution–pooling layers followed by fully connected (FC) layers, taking a 1×103 feature vector as input [17], [21] (Fig. 3). To address the complexity of dual activation mechanisms, we

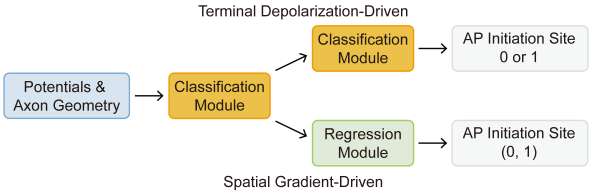


Fig. 2. The hierarchical prediction strategy for AP initiation site estimation.

implemented a hierarchical prediction strategy rather than a single monolithic network (Fig. 2). Specifically, an initial classification network identifies the activation mechanism (spatial gradient-driven vs. terminal depolarization-driven). Based on this classification, the task bifurcates: for terminal depolarization–driven activation cases, a sub-classification network determines whether the AP originated from the proximal (0-end) or distal (1-end) terminal node; for spatial gradient-driven cases, a regression network was employed to predict the relative initiation location along the axonal trajectory.

All neural networks were implemented, trained, and evaluated using TensorFlow [22]. The rectified linear unit (ReLU) was used as the activation function for all fully connected layers in the CNN. For classification tasks, a sigmoid activation function was applied to the output layer. Hyperparameter optimization was performed to identify the optimal network configurations. The number of convolution–pooling layers ranged from three to six, while the number of fully connected layers ranged from two to four. The learning rate was set to 0.0003. The dataset was randomly split into a training set (70%) and a test set (30%). To mitigate overfitting, a ten-fold cross-validation strategy was applied to the training set, where the training data were divided into ten subsets, and in each fold, one subset was used for validation while the remaining subsets were used for training. The training process was terminated if there was no improvement in validation performance over 30 epochs or if the maximum number of iterations had been reached. The model weights that achieved the best performance on the validation set were saved, resulting in ten trained models. The final prediction was obtained by averaging the outputs of these models using an ensemble approach [23], which effectively improves robustness and reduces overfitting. Classification accuracy was used as the performance metric for classification tasks, while mean squared error (MSE) was employed for regression tasks. The trained models were subsequently validated on the test set using metrics including classification accuracy, Mean Absolute Error (MAE), MSE, and R^2.

Crucially, the test set remained completely disjoint from the training and validation processes to strictly preclude information leakage. First and foremost, a substantial portion of the dataset consists of stochastically generated samples that are entirely independent of one another, which inherently eliminates the potential for data leakage. To further ensure the model's generalization, we implemented a rigorous data partitioning and randomization protocol for the remaining samples. Regarding the remaining data, although certain entries in the training and test sets might share electric field profiles originating from the same FEM model, every individual instance was independently and stochastically assigned a unique combination of stimulation pulse width, fiber diameter, and axonal length. This high-dimensional parameter randomization prevents the network from relying on memorized field patterns. Furthermore, even under identical global stimulation settings, the inherent variations in axonal trajectories within non-uniform electric fields yield distinct extracellular potential distributions. This spatial diversity provides an additional layer of protection against over-fitting to specific electrode configurations, ensuring that the model learns the underlying biophysical mechanisms rather than dataset-specific artifacts.

C. Prediction of AP Initiation Nodes Using the Activating Function Method

The activating function (AF) method has been widely used to infer the initiation site of action potentials [12], [13]. Rattay demonstrated that the second spatial derivative of the extracellular potential along the axon plays a critical role in action potential generation and proposed it as the activating function [9]. A positive activating function facilitates action potential initiation, whereas a negative activating function suppresses it. Consequently, the node with the maximum AF value is commonly regarded as the AP initiation node.

Although this approach captures the most relevant physical quantity governing neural excitation, it does not account for the contributions of neighboring nodes to action potential generation and is not applicable to terminal depolarization–driven activation. Nevertheless, we applied the AF method to predict AP initiation nodes for spatial gradient–driven samples in the test set. Specifically, the second spatial derivative of the extracellular potential was computed using a finite-difference scheme, and the AF values at individual nodes were obtained via interpolation. The node with the maximum AF value was identified as the predicted AP initiation node. A subset of these samples exhibited constant second spatial derivatives, rendering the AF method inapplicable. These samples were therefore excluded from the analysis, resulting in a total of 28,516 samples retained for AF-based prediction.

D. Prediction of AP Initiation Nodes Using the Driving Force Method

To incorporate the influence of neighboring nodes on action potential generation, the driving force (DF) method has been proposed as an extension of the AF approach. In this method, the activating functions of multiple adjacent nodes are combined through weighted summation to estimate both the AP initiation site and the activation threshold [14]–[16]. While the DF method has been shown to improve prediction performance to some extent, the node-specific weights depend on axonal geometric parameters and stimulus pulse width, and the method is difficult to generalize to complex stimulation scenarios.

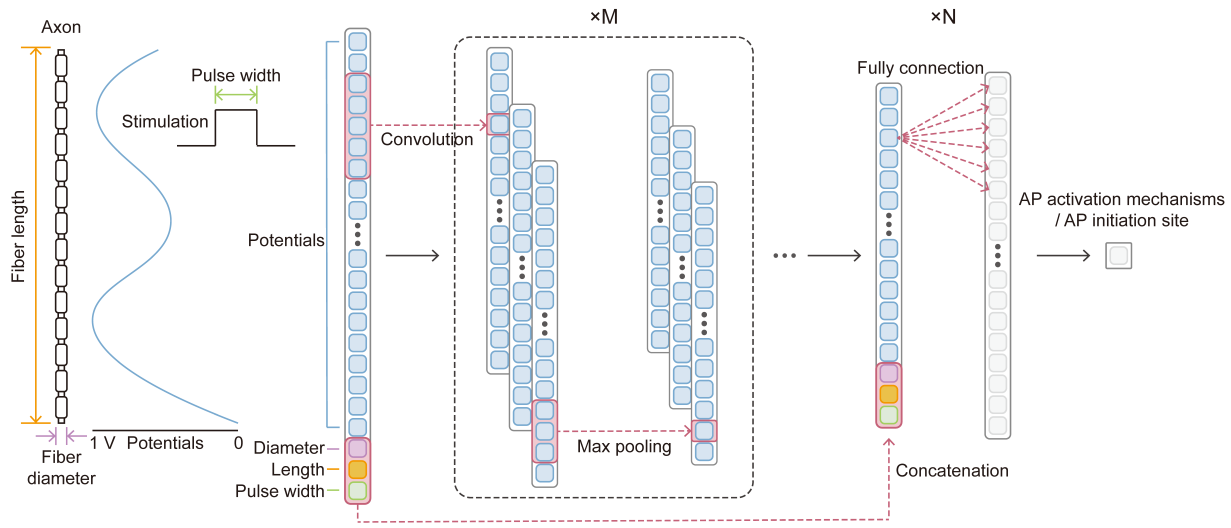


Fig. 3. Neural network structures.

In this study, we applied the 11-node DF method to predict the AP initiation nodes for the same set of 28,516 samples. The driving force at each node was computed using weighting coefficients reported in the literature [15]. For axon diameters and pulse widths not explicitly covered in the reported data, two-dimensional interpolation was employed to estimate the corresponding weights. The node with the maximum DF value was then selected as the predicted AP initiation node [16].

III. Results

A. CNN Accurately Identifies AP Activation Mechanisms

The proposed CNN demonstrated a strong and reliable capability to classify action potential (AP) activation mechanisms. The classification accuracy reached 98.93% on the training set and remained consistently high at 98.46% on the independent test set. The corresponding confusion matrix is presented in Fig. 4. The consistently high accuracy across both datasets indicates that the CNN generalizes well and does not exhibit significant overfitting.

This high classification performance suggests that the CNN effectively integrates the spatial characteristics of extracellular potential distributions with axonal geometric features, enabling accurate inference of the underlying membrane response patterns. In contrast to conventional activating function (AF)–based and driving force (DF)–based approaches, which rely on explicit analytical formulations and simplifying assumptions, the proposed CNN directly infers the activation mechanism from the extracellular potential profile. This data-driven strategy not only avoids restrictive assumptions but also substantially reduces the computational burden associated with numerical differentiation and node-wise evaluation.

B. CNN Accurately Predicts AP Initiation Sites for Terminal Depolarization–Driven Activation

For terminal depolarization–driven activation, the CNN exhibited excellent performance in identifying both the AP-initiating terminal and the corresponding direction of propagation. The classification accuracy for AP initiation terminals reached 99.81% on the training set and 99.59% on the test set. The associated confusion matrix is shown in Fig. 5. The minimal performance gap between the training and test sets further demonstrates the robustness and stability of the proposed model.

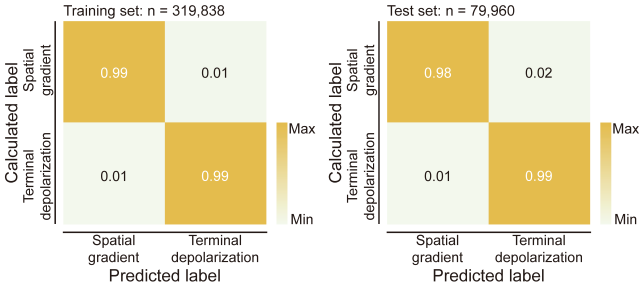


Fig. 4. Confusion matrix between predicted and calculated AP activation mechanisms.

These results indicate that the CNN can reliably capture the characteristic features of terminal depolarization–driven activation, even across a wide range of axonal diameters, lengths, and stimulation pulse widths. Importantly, this task cannot be addressed by AF- or DF-based methods, highlighting the unique advantage of the CNN-based framework in handling terminally initiated APs.

C. CNN Reliably Predicts AP Initiation Sites for Spatial Gradient–Driven Activation

For action potentials initiated at internal axonal nodes, corresponding to spatial gradient–driven activation, the CNN achieved high prediction accuracy. On the training set, the mean squared error (MSE) was 6.98×10^{-4}, the mean absolute error (MAE) was 1.09×10^{-2}, and the coefficient of

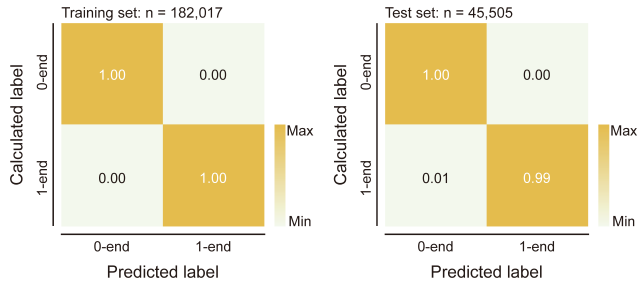


Fig. 5. Confusion matrix between predicted and calculated AP initiation terminals.

determination (R^2) reached 0.985. On the test set, the MSE was 1.10×10^{-3}, the MAE was 1.21×10^{-2}, and the R^2 value remained high at 0.976, indicating strong generalization performance.

For the test set, the node closest to the CNN-predicted normalized initiation location was identified as the predicted AP initiation node. The comparison between the predicted and ground-truth initiation sites is illustrated in the scatter plot shown in Fig. 6a. For comparison, Fig. 6b and Fig. 6c present the corresponding scatter plots obtained using the AF and DF methods, respectively. As shown in these figures, the CNN-based approach is applicable to a broader range of scenarios, including cases in which the activating function is spatially constant, where AF-based methods inherently fail. Moreover, the CNN consistently achieves substantially higher prediction accuracy across the entire dataset.

To further quantify prediction performance, we computed the absolute error between the predicted and true initiation nodes for all three methods in the test set (Fig. 6d–g). The CNN achieved an average error of 0.72 nodes, whereas the AF method exhibited the largest error with an average of 6.78 nodes. The DF method achieved an intermediate performance, with an average error of 3.27 nodes. Using the CNN-based approach, 98.1% of the samples exhibited a prediction error within five nodes. In contrast, only 60.5% of the samples predicted using the AF method achieved errors within five nodes, while the corresponding proportion for the DF method was 82.6%.

Recognizing that the AF method is inherently limited in cases with constant second spatial derivatives, we conducted a secondary evaluation on a matched subset ($n = 28,516$) that excluded such instances to ensure a fair comparison. On this subset, the CNN maintained its superior performance with an average error of 0.80 nodes and a 97.8% success rate within a five-node error margin. These results clearly demonstrate that the CNN-based approach significantly outperforms both AF- and DF-based methods in predicting AP initiation sites.

IV. Conclusions

In this study, we proposed a data-driven framework based on convolutional neural networks (CNNs) to predict both the activation mechanism and the initiation site of action potentials (APs) in myelinated axons under extracellular electrical stimulation. Using a large-scale dataset generated from the MRG axon model, the proposed approach accurately distinguished between spatial gradient–driven and terminal depolarization–driven activation mechanisms, and subsequently achieved highly accurate predictions of AP initiation sites for each mechanism. Compared with conventional activating function (AF)–based and driving force (DF)–based methods, the CNN-based framework demonstrated substantially improved prediction accuracy and broader applicability across diverse stimulation and axonal parameter settings.

The proposed CNN framework offers a transformative leap in computational throughput compared to conventional biophysical solvers. Benchmarking on a single CPU core, the gold-standard NEURON environment requires over 48 hours to simulate the membrane potential dynamics for a batch of 100,000 axons (each 10 cm in length and 5.7 μm in diameter). In contrast, our trained CNN surrogate completes the inference for the same dataset in less than 1 minute [17]. This thousand-fold acceleration facilitates near-instantaneous prediction, rendering the framework highly suitable for large-scale parameter sweeps and real-time optimization in neuromodulation applications.

A key feature of the proposed approach is that it directly uses the extracellular potential distribution as input, rather than relying on derived quantities such as spatial second-order derivatives [21], [24]. In our previous work, it was demonstrated that the activation threshold of axons can be accurately predicted using the potential distribution [17]. By preserving the full spatial structure of the extracellular field, the CNN is able to exploit information that is discarded in traditional analytical formulations [25], including the collective influence of neighboring nodes and boundary effects at axonal terminals. This enables robust prediction not only for spatial gradient–driven activation, but also for terminal depolarization–driven activation, which cannot be addressed by AF- or DF-based methods. Furthermore, the proposed hierarchical prediction strategy effectively accounts for the coexistence of distinct activation mechanisms, thereby avoiding the ambiguity and performance degradation observed when attempting to directly predict AP initiation sites with a single regression model.

Despite these advantages, several limitations should be acknowledged. First, the current framework was trained and evaluated using simulation data derived from the MRG axon model under single-pulse stimulation, and its generalizability to other axonal models, stimulation waveforms, or pathological conditions remains to be investigated. Second, while the CNN provides accurate predictions, its internal representations are not yet fully interpretable in biophysical terms, and further work is needed to establish stronger links between learned features and known electrophysiological mechanisms.

Future work will focus on extending the proposed framework to more complex stimulation paradigms, including time-varying waveforms and three-dimensional extracellular fields. In addition, incorporating explainable artificial intelligence

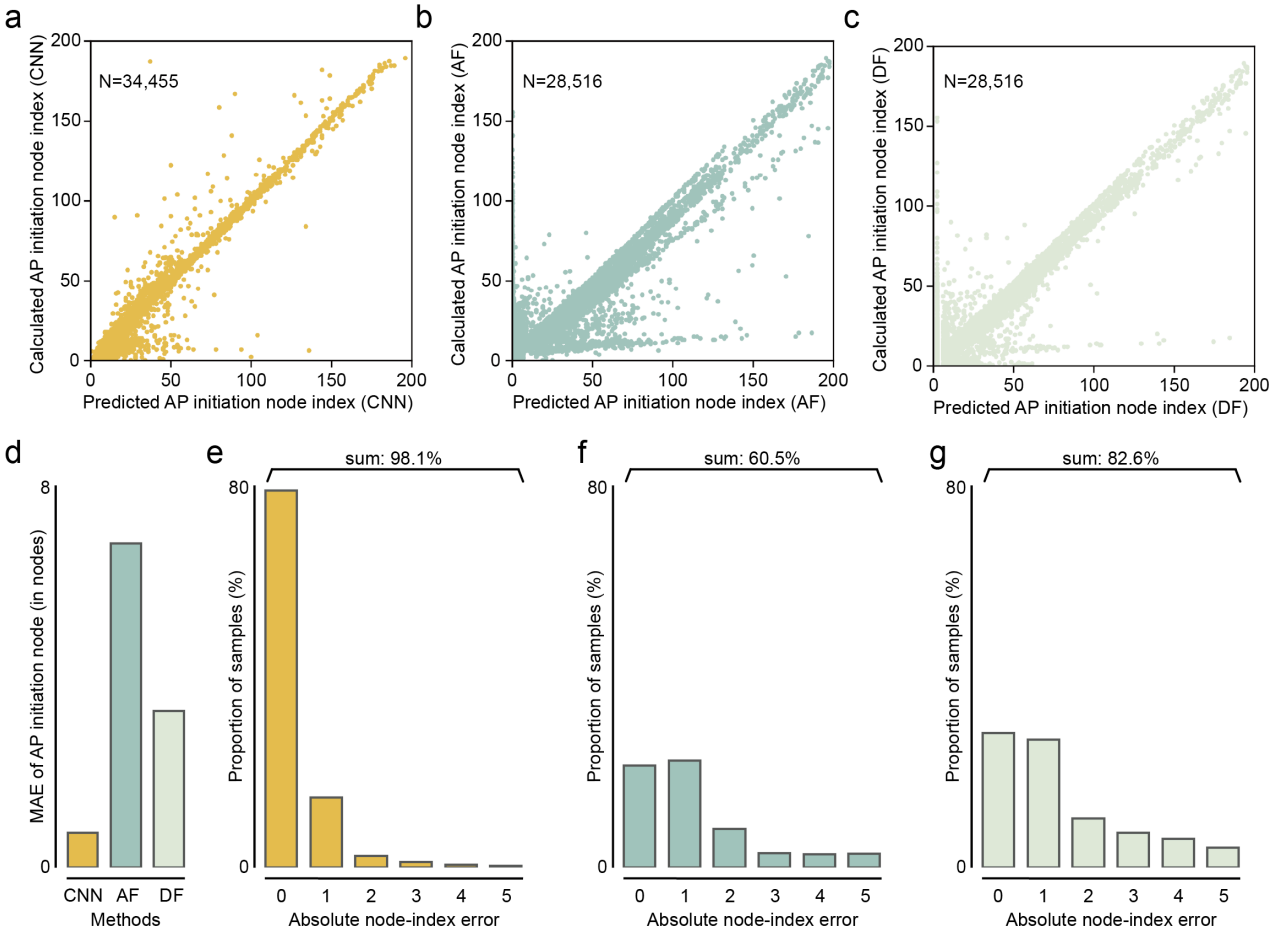


Fig. 6. Predictive performance of AP initiation nodes under spatial gradient-driven conditions. (a)-(c), The comparison between predicted and calculated initiation nodes using three methods. (d), Mean absolute node error between predicted and calculated AP initiation nodes. (e)-(g), Absolute node error distributions between predicted and calculated initiation nodes.

techniques may help to further elucidate the relationship between extracellular field characteristics and AP initiation dynamics. Finally, integrating the proposed approach into patient-specific neuromodulation modeling pipelines may enable rapid and accurate estimation of neural activation patterns, facilitating stimulation optimization and personalized therapy design.

Overall, this work demonstrates that deep learning–based models can effectively capture the complex and multimodal relationship between extracellular electrical stimulation and axonal responses. By providing a fast, accurate, and flexible alternative to conventional analytical methods, the proposed framework offers a promising tool for large-scale neural activation prediction and has potential applications in the design and optimization of neuromodulation therapies.

References

[1] S. Harkema, Y. Gerasimenko, J. Hodes, J. Burdick, C. Angeli, Y. Chen, C. Ferreira, A. Willhite, E. Rejc, R. G. Grossman, *et al.*, "Effect of epidural stimulation of the lumbosacral spinal cord on voluntary movement, standing, and assisted stepping after motor complete paraplegia: a case study," *The Lancet*, vol. 377, no. 9781, pp. 1938–1947, 2011.

[2] F. B. Wagner, J.-B. Mignardot, L. Goff-Mignardot, G. Camille, R. Demesmaeker, S. Komi, M. Capogrosso, A. Rowald, I. Seáñez, M. Caban, *et al.*, "Targeted neurotechnology restores walking in humans with spinal cord injury," *Nature*, vol. 563, no. 7729, pp. 65–71, 2018.

[3] A. Rowald, S. Komi, R. Demesmaeker, E. Baaklini, S. D. Hernandez-Charpak, E. Paoles, H. Montanaro, A. Cassara, F. Becce, B. Lloyd, *et al.*, "Activity-dependent spinal cord neuromodulation rapidly restores trunk and leg motor functions after complete paralysis," *Nature medicine*, vol. 28, no. 2, pp. 260–271, 2022.

[4] J. S. Perlmutter and J. W. Mink, "Deep brain stimulation," *Annu. Rev. Neurosci.*, vol. 29, no. 1, pp. 229–257, 2006.

[5] K. Chakravarthy, A. Nava, P. J. Christo, and K. Williams, "Review of recent advances in peripheral nerve stimulation (pns)," *Current pain and headache reports*, vol. 20, no. 11, p. 60, 2016.

[6] C. C. McIntyre and W. M. Grill, "Extracellular stimulation of central neurons: influence of stimulus waveform and frequency on neuronal output," *Journal of neurophysiology*, vol. 88, no. 4, pp. 1592–1604, 2002.

[7] C. C. McIntyre, A. G. Richardson, and W. M. Grill, "Modeling the excitability of mammalian nerve fibers: influence of afterpotentials on the recovery cycle," *Journal of neurophysiology*, vol. 87, no. 2, pp. 995–1006, 2002.

[8] J. P. Reilly, "Survey of numerical electrostimulation models," *Physics in Medicine & Biology*, vol. 61, no. 12, p. 4346, 2016.

[9] F. Rattay, "Analysis of models for external stimulation of axons," *IEEE transactions on biomedical engineering*, no. 10, pp. 974–977, 1986.

[10] F. Rattay, S. Resatz, P. Lutter, K. Minassian, B. Jilge, and M. Dimitrijevic, "Mechanisms of electrical stimulation with neural prostheses,"

Neuromodulation: Technology at the Neural Interface, vol. 6, no. 1, pp. 42–56, 2003.

[11] M. L. Hines and N. T. Carnevale, "The neuron simulation environment," *Neural computation*, vol. 9, no. 6, pp. 1179–1209, 1997.

[12] C. R. Butson, S. E. Cooper, J. M. Henderson, and C. C. Mcintyre, "Patient-specific analysis of the volume of tissue activated during deep brain stimulation.," *Neuroimage*, vol. 34, no. 2, pp. 661–670, 2007.

[13] M. Åström, E. Diczfalusy, H. Martens, and K. Wårdell, "Relationship between neural activation and electric field distribution during deep brain stimulation," *IEEE Transactions on Biomedical Engineering*, vol. 62, no. 2, pp. 664–672, 2014.

[14] E. N. Warman, W. M. Grill, and D. Durand, "Modeling the effects of electric fields on nerve fibers: determination of excitation thresholds," *IEEE Transactions on Biomedical Engineering*, vol. 39, no. 12, pp. 1244–1254, 1992.

[15] E. J. Peterson, O. Izad, and D. J. Tyler, "Predicting myelinated axon activation using spatial characteristics of the extracellular field," *Journal of neural engineering*, vol. 8, no. 4, p. 046030, 2011.

[16] B. Howell, K. Gunalan, and C. C. McIntyre, "A driving-force predictor for estimating pathway activation in patient-specific models of deep brain stimulation," *Neuromodulation: Technology at the Neural Interface*, vol. 22, no. 4, pp. 403–415, 2019.

[17] H. Li, S. Wang, X. Luo, C. Jiang, and B. Zhang, "Versatile neural activation predictor with axon structure tailoring capability enabling personalized neuromodulation computation," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 33, pp. 3986–3997, 2025.

[18] X. Zhang, Y. Lu, B. Zhang, H. Li, Q. Yao, J. Lim, Y. Wei, K. He, C. Zuo, Y. Sui, *et al.*, "Voluntary walking related joint movement training with targeted epidural electrical stimulation enabled neural recovery for individuals with spinal cord injury," *Science bulletin*, vol. 69, no. 22, pp. 3507–3511, 2024.

[19] H. Li, J. Shi, Y. Sui, and B. Zhang, "Modeling and simulation of spatial characteristics in epidural electrical stimulation for targeted neural activation," in *2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pp. 1–7, 2025.

[20] M. L. Hines, T. Morse, M. Migliore, N. T. Carnevale, and G. M. Shepherd, "Modeldb: a database to support computational neuroscience," *Journal of computational neuroscience*, vol. 17, pp. 7–11, 2004.

[21] S. Wang, R. Ma, Q. Yuan, H. Li, and C. Jiang, "Efficient, robust, and accurate cnn predictor for neuronal activation in directional deep brain stimulation," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 33, pp. 1685–1694, 2025.

[22] M. Abadi, P. Barham, J. Chen, Z. Chen, A. Davis, J. Dean, M. Devin, S. Ghemawat, G. Irving, M. Isard, *et al.*, "{TensorFlow}: a system for {Large-Scale} machine learning," in *12th USENIX symposium on operating systems design and implementation (OSDI 16)*, pp. 265–283, 2016.

[23] T. G. Dietterich, "Ensemble methods in machine learning," in *International workshop on multiple classifier systems*, pp. 1–15, Springer, 2000.

[24] J. Golabek, M. Schiefer, J. K. Wong, S. Saxena, and E. Patrick, "Artificial neural network-based rapid predictor of biological nerve fiber activation for dbs applications," *Journal of Neural Engineering*, vol. 20, no. 1, p. 016001, 2023.

[25] S. Shen, T. Shao, K. Zhou, C. Jiang, F. Luo, and Y. Yang, "Second-order neural network training using complex-step directional derivative," *arXiv preprint arXiv:2009.07098*, 2020.